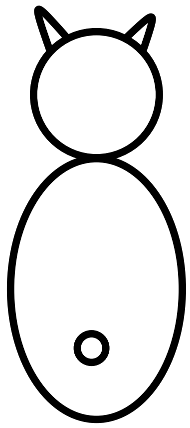
Summer Examinations 2021-22

Exam Code(s)	3BA1, 4BA1, 4BME1, 1OA1, 3BMS2, 3BS91, 3FM2, 1HA1
Exam	Third & Fourth Year & HDip
Module	TOPOLOGY
Module Codes	MA342 & MA535
External Examiner(s)	Prof C. Roney-Dougal
Internal Examiner(s)	Prof G. Ellis*
	Dr N. Madden
Instructions	Attempt **all** questions.
	There are ten questions, each carrying 10 marks.
Duration	2 hours
No. of Pages	4 pages (including this cover page)
Discipline	Mathematics

Requirements:

Release in Exam Venue	Yes ☑	No ☐
Release to Library	Yes ☑	No ☐
Non-programmable calculator	Yes ☐	No ☑
Mathematical Tables	Yes ☐	No ☑

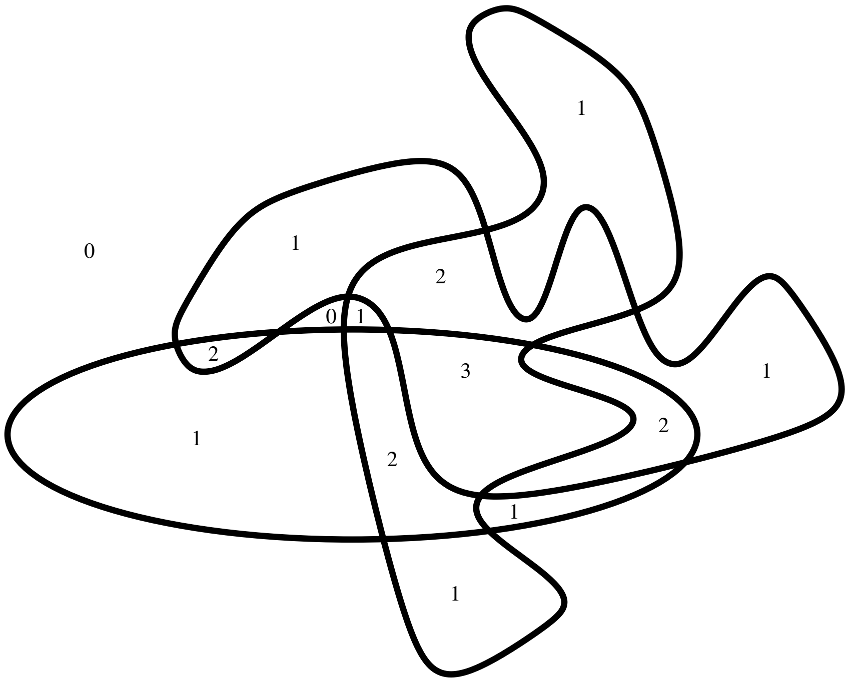
1. The digital image



represents a closed subspace $X \subset \mathbb{E}^2$ formed as a union of many unit squares of the form $[m, m+1] \times [n, n+1]$ for integers m, n. Each such square corresponds to a tiny black pixel and is understood to comprise four vertices, four edges and one 2-dimensional face. The space X has 2 connected components.

Define the Euler characteristic $\chi(X)$. Then calculate $\chi(X)$, giving details of your calculation.

2. The following picture shows the boundaries of several closed subspaces $U_1, \ldots, U_t \subseteq \mathbb{E}^2$ of common Euler characteristic $\chi(U_i) = 1$. No two boundaries are tangential.



The numbers in the interiors of the subspaces and their intersections represent the weight function $\omega\colon X \to \mathbb{Z}$ where $X = U_1 \cup \ldots \cup U_t$ and $\omega(x) = |\{i : x \in U_i\}|$. Define the Euler integral

$$\int_X \omega \, d\chi \quad .$$

Then calculate this Euler integral, showing your calculations. Derive the value of t from your calculation and state any formula you use in your derivation.

3. Explain what is meant by a triangulation of a topological space.

Take the unit square $X = [0,1] \times [0,1]$ in $\mathbb{E}^2$, with the subspace topology, and partition X into the following sets:

(a) sets consisting of pairs of points $(x, 0)$ and $(x, 1)$, where $0 < x < 1$.
(b) sets consisting of pairs of points $(0, y)$ and $(1, y)$, where $0 < y < 1$.
(c) sets consisting of a single point (x, y), where $0 < x < 1$ and $0 < y < 1$.
(d) The set consisting of the four points $(0,0), (0,1), (1,0), (1,1)$.

Let Y be the associated identification space. Exhibit a triangulation for Y. Then determine the Euler characteristic $\chi(Y)$.

4. Explain what it means for a topological space to be be *Hausdorff*.

For each of the following sets X and collections T of *open* subsets decide if the pair X, T satisfies the axioms of a topological space. If it does, determine whether the topological space is Hausdorff. If the axioms are not satisfied then exhibit one axiom that fails.

(a) $X = \{1,2,3,4,5,6\}$ and $T = \{\emptyset, \{1\}, \{1,2\}, \{1,3,4,5\}, \{2,3,4,5\}, \{1,2,3,4,5,6\}\}$.
(b) $X = \mathbb{Q}$ and a subset $U \subset \mathbb{Q}$ is *open* if and only if $U = \mathbb{Q}$ or U is finite.
(c) $X = \mathbb{Z}$ and a subset $U \subset \mathbb{Z}$ is *open* if and only if $3 \notin U$ or $U = \mathbb{Z}$.
(d) $X = \mathbb{Z}$ and a subset $U \subset \mathbb{Z}$ is *open* if and only if it has the property that $u \in U$ implies $-u \in U$.

5. Let the set $\mathbb{Z}^2 = \{(m,n) \ : \ m, n \in \mathbb{Z}\}$ be endowed with the cofinite topology (in which a set is *open* if and only if it is empty or its complement is finite).

(a) Define what it means for a topological space to be *connected*. Is $\mathbb{Z}^2$ connected with the given topology? Justify your answer.
(b) Exhibit a function $g \colon \mathbb{Z}^2 \to \mathbb{Z}^2$ that is not continuous with this topology, and explain why it is not continuous.

6. Consider the real line $\mathbb{R}$ endowed with the usual topology. Let $\mathbb{Z}$ denote the subspace of integers. Let $\mathbb{Q}$ denote the subspace of rational numbers. Is $\mathbb{Z}$ homeomorphic to $\mathbb{Q}$? Provide a justification of your answer.

7. Consider the following seven letters

A B C D E F G

as seven connected subspaces of the Euclidean plane $\mathbb{E}^2$.

(a) Partition the letters into sets with all letters in a set homeomorphic to each other and with no two letters from distinct sets homeomorphic.
(b) Partition the letters into sets with all letters in a set homotopy equivalent to each other and with no two letters from distinct sets homotopy equivalent.

8. Answer **either** (a) **or** (b). Pay particular attention to your syle of exposition.

(a) State the Brouwer Fixed-Point Theorem and use it to prove that any square matrix with positive real entries has a positive real eigenvalue with corresponding eigenvector having only non-negative real entries.

(a) Deduce from the bijection $[S^1, S^1] \cong \mathbb{Z}$ that any polynomial $a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ with coefficients $a_i \in \mathbb{C}$ has at least one zero in $\mathbb{C}$.

9. Answer **either** (a) **or** (b). Pay particular attention to your style of exposition.

(a) Use the fact that the Euler characteristic of a triangulable space is a homotopy invariant to prove the Brouwer Fixed Point Theorem.

(b) Prove that a compact subset of a Hausdorff topological space is closed.

10. Answer **either** (a) **or** (b). Pay particular attention to your syle of exposition.

(a) Describe a surjective continuous map $f\colon [0,1] \to \Delta$ where $\Delta \subset \mathbb{R}^2$ is a solid equilateral triangle. (Do not prove surjectivity of the map.)

(b) Describe what is meant by a *Nash Equilibrium*, explaining any concepts from Game Theory that you use.